

Department of Electrical and Computer Engineering
University of Wisconsin – Madison
ECE 552 Introductions to Computer Architecture
In-Class Exercise (10/17)

- Given a processor with 32-bit memory addresses, how many **set index bits** are needed for a **4KB (4096B) direct-mapped** cache with **32-byte blocks**?

Answer:

$$\log_2(4096 / 32) = 7 \text{ bits}$$

- Consider a **32B direct-mapped** cache with **8-byte blocks**. Assume addresses are 8-bit, and memory is byte-addressable. Fill in the blanks in the table below for the given sequence of loads; loads are performed sequentially from top to bottom. For each load in the table, specify the **tag** bits, **set index** bits, **tags** in the cache **after** the access, and whether the load is a **hit (H)** / **miss (M)**. The first two rows are provided for you.

Address (hex)	Tag (binary)	Index (binary)	Tags in the cache after the load (binary)				H/M
			Set 0	Set 1	Set 2	Set 3	
0x38	001	11	000	001	111	001	H
0xC2	110	00	110	001	111	001	M
0x14	000	10	110	001	000	001	M
0xA9	101	01	110	101	000	001	M
0xC3	110	00	110	101	000	001	H
0x7B	011	11	110	101	000	011	M
0xC4	110	00	110	101	000	011	H
0x11	000	10	110	101	000	011	H
0xAA	101	01	110	101	000	011	H
0x67	011	00	011	101	000	011	M
0xC5	110	00	110	101	000	011	M
0x39	001	11	110	101	000	001	M

3. Consider a processor with a **data cache** and **instruction cache**, both **direct-mapped** with **64-byte blocks** (16 words per block). You run the following MIPS program. Assume that the first instruction is at address 0x00000100 in memory, and both caches are **initially empty**:

```

                addi  $t0, $zero, 3
LOOP0:         addi  $t1, $zero, 1024
LOOP1:         lw    $t2, 0($t1)
                sub   $s7, $s7, $t2
                addi  $t1, $t1, 4
                slti  $t3, $t1, 2048
                bne   $t3, $zero, LOOP1
                addi  $t0, $t0, -1
                bne   $t0, $zero, LOOP0

```

- a) Given **256B** caches, how many **data cache misses** would this program yield?

Answer:

$$\frac{2048 - 1024}{4} * \frac{1}{16} * 3 = 48 \text{ misses}$$

Inner loop goes from 1024 through 2048 by increments of 4. Cache block has 16 words, so the first fetch is a miss but 16 words are read. The next 15 are hits. This question says cache is 256B, which is $\frac{256}{64} = 4$ cache blocks, so cache block reads 9-3 are compulsory misses and every cache block read after that is a capacity eviction. Multiply by 3 for the outer loop.

- b) Given **256B** caches, how many **data cache hits** would this program yield?

Answer:

$$\frac{2048 - 1024}{4} * \frac{15}{16} * 3 = 720 \text{ hits}$$

Same as above, but counting the 15 spatial hits per cache line.

- c) Given **infinite-size** caches, how many **data cache misses** would this program yield?

Answer:

$$\frac{2048 - 1024}{4} * \frac{1}{16} = 16 \text{ misses}$$

Infinite-size cache means there are infinite cache blocks. There's still a miss for the 0th word of each cache block, but nothing is evicted so the outer loop doesn't produce new misses.

- d) Given **infinite-size** caches, how many **data cache hits** would this program yield?

Answer:

$$\frac{2048 - 1024}{4} * 3 - 16 \text{ misses} = 752 \text{ hits}$$

- e) Given **infinite-size** caches, how many **inst cache misses** would this program yield?

Answer:

The program is 9 instructions. Instructions are 1 word each. The entire program fits inside the 16 word cache block starting at address 0x100 (a cache block address boundary), so there's only 1 miss on the first instruction.